



Human Perception



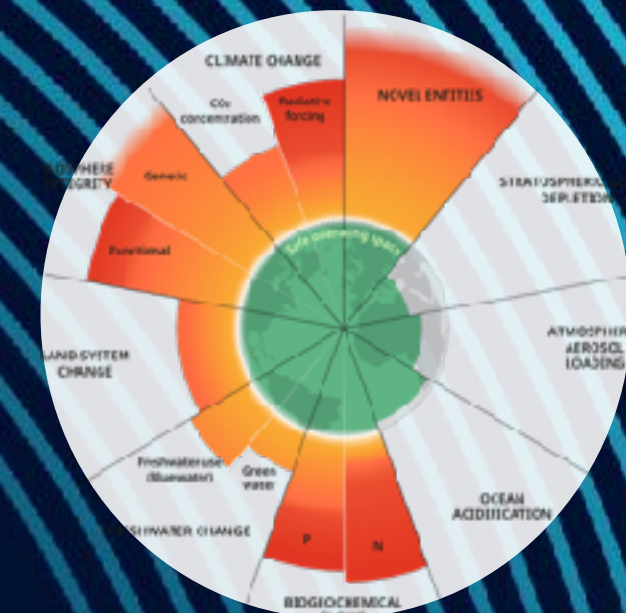
Cognitive Bias



Seeing is Believing



Ethics in AI Imaging



Sufficiency for Image Science



Digital footprint

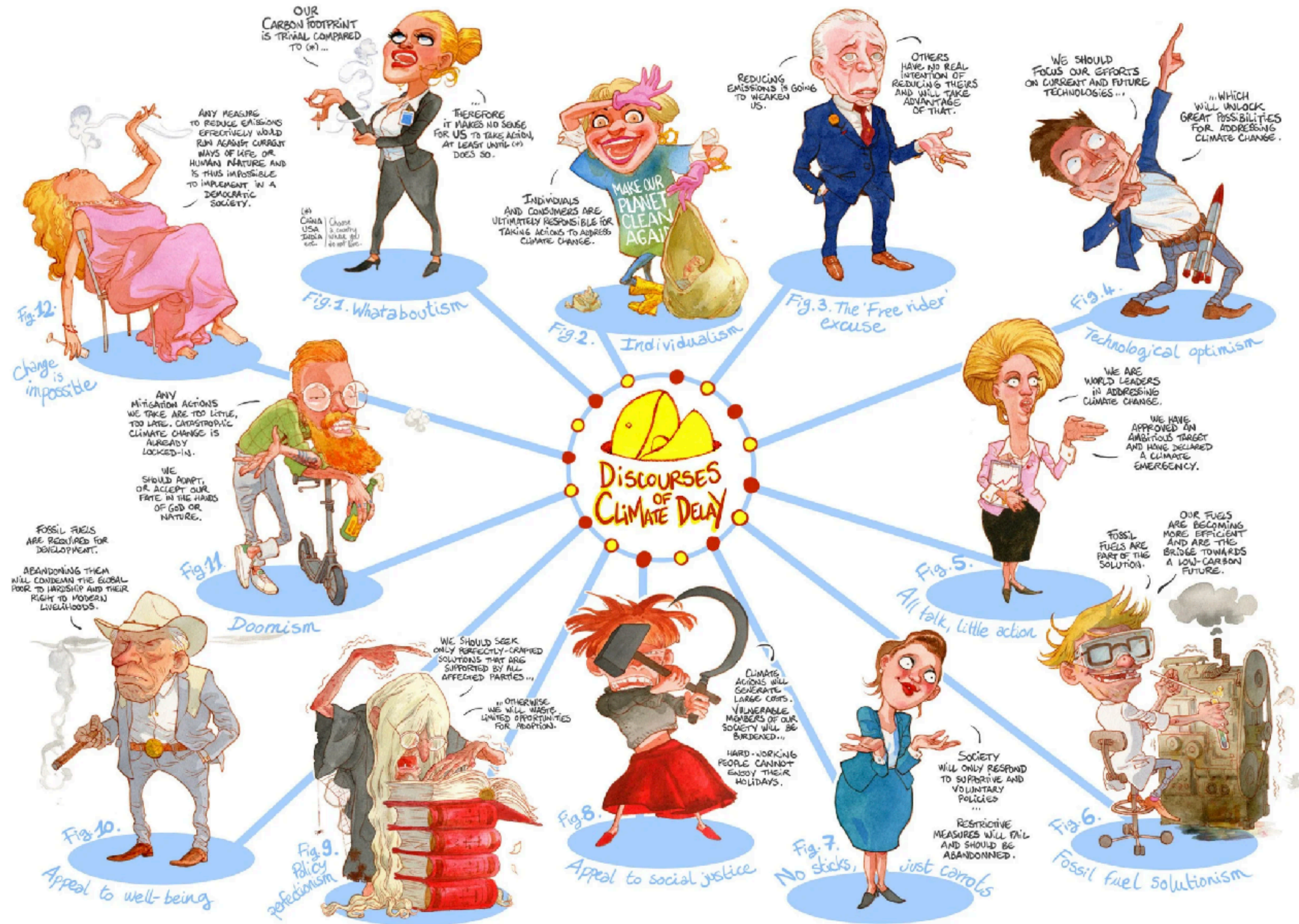


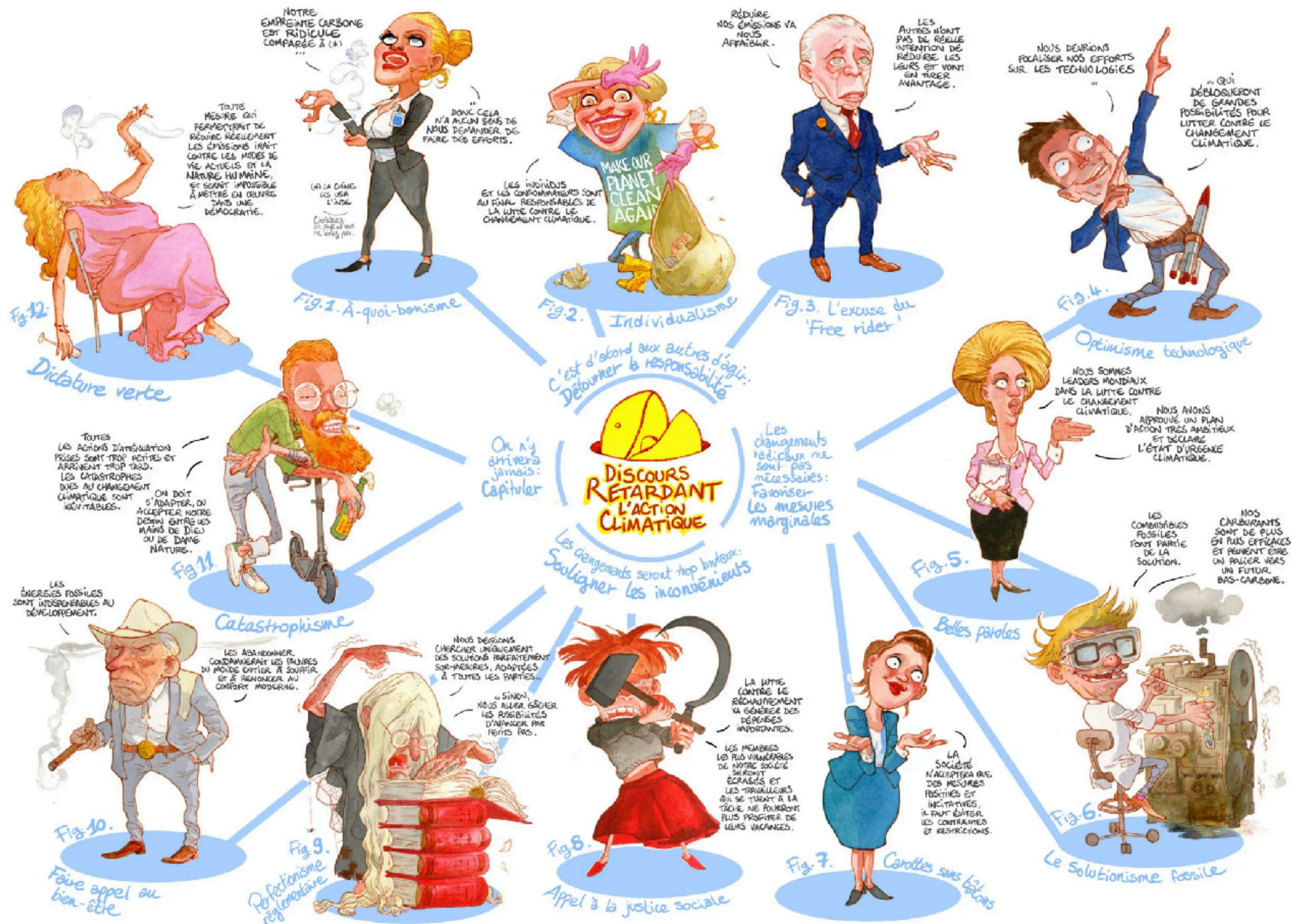
Open
Science

Responsible AI refers to the development, deployment, and use of AI systems in a way that aligns with ethical principles and societal values.

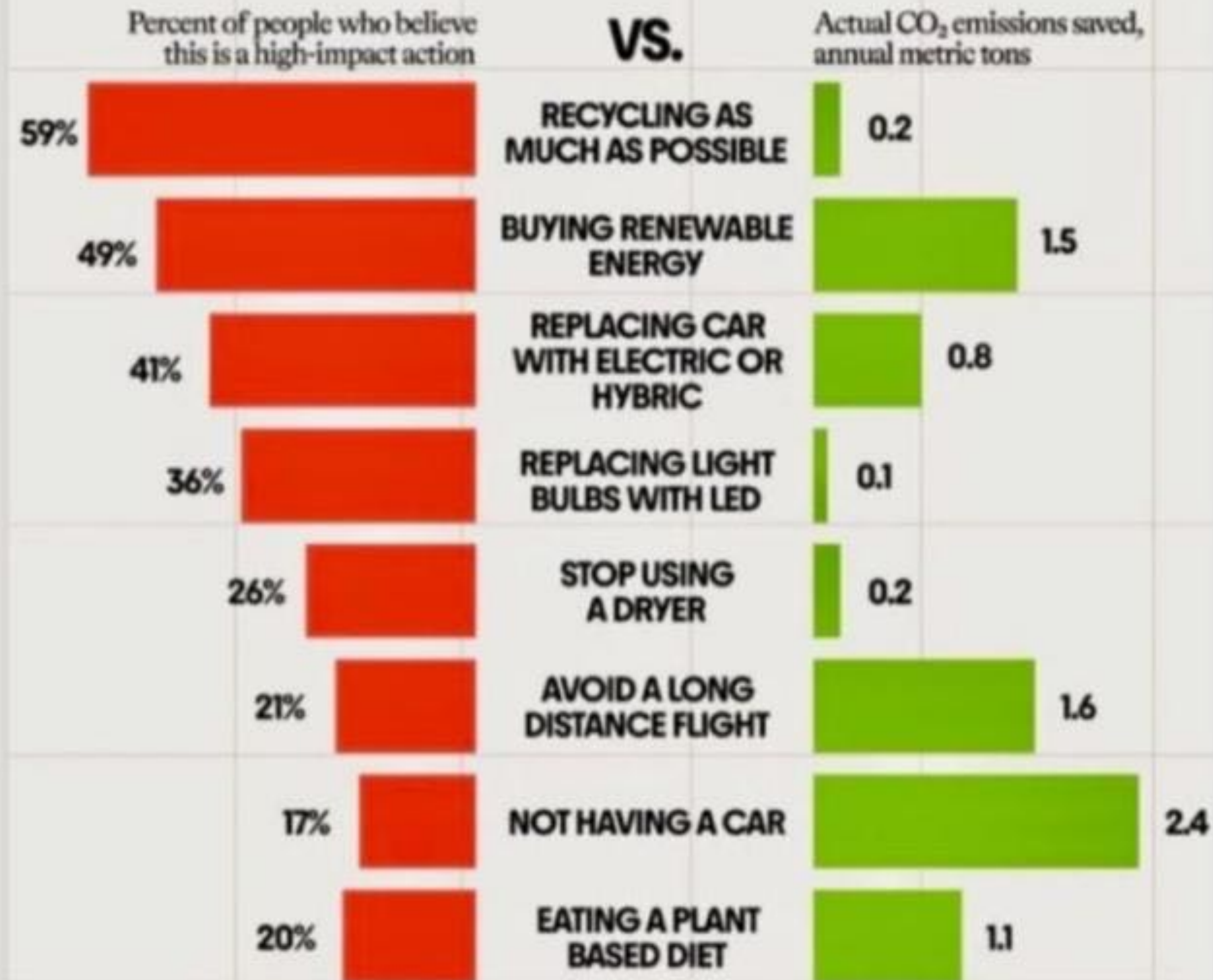


<https://www.leolinne.com/-discourses-of-climate-delay/english>





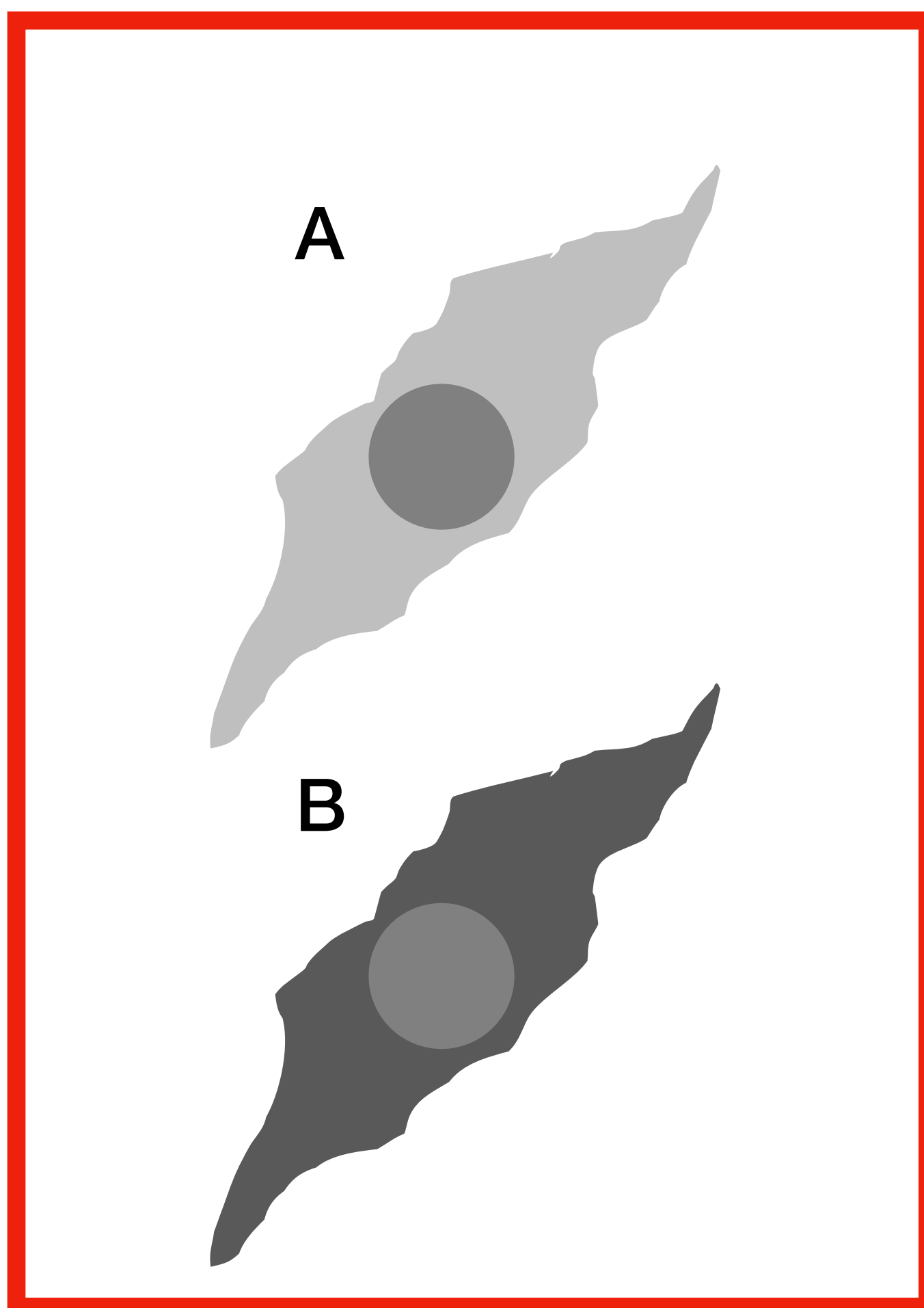
CARBON FOOTPRINT MISCONCEPTIONS



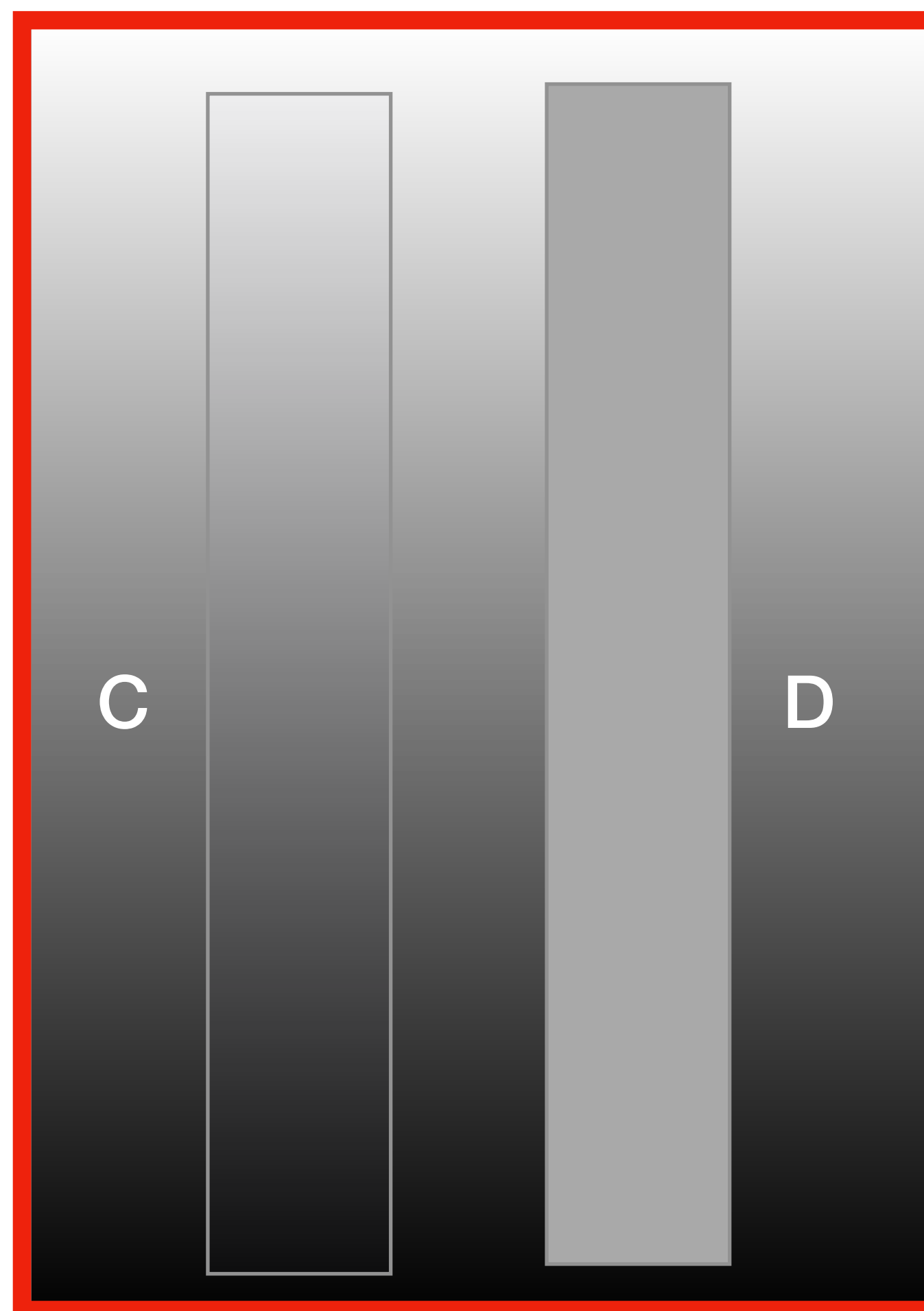


Human Perception

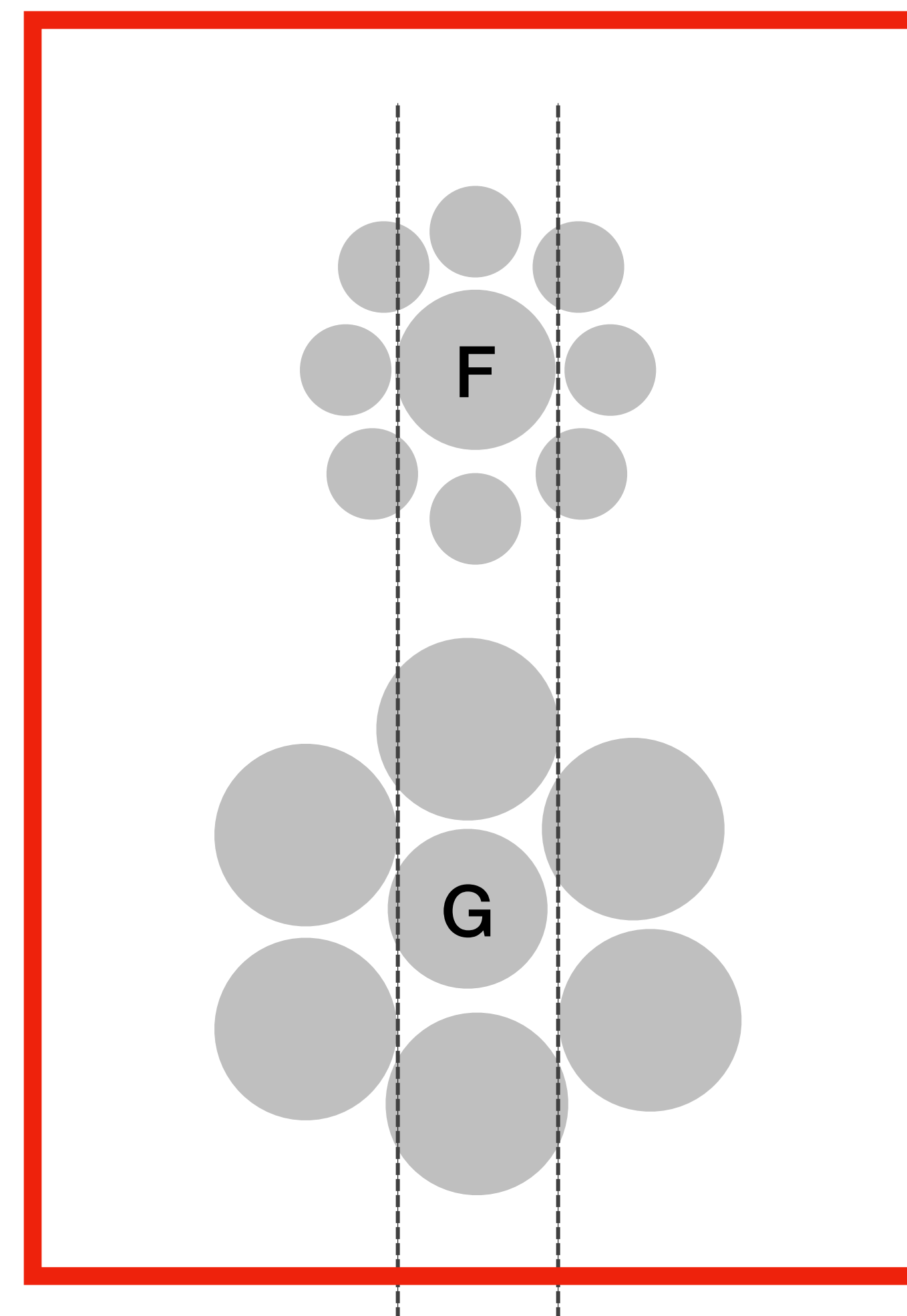
Which disk is brighter?



Which bar has a gradient?

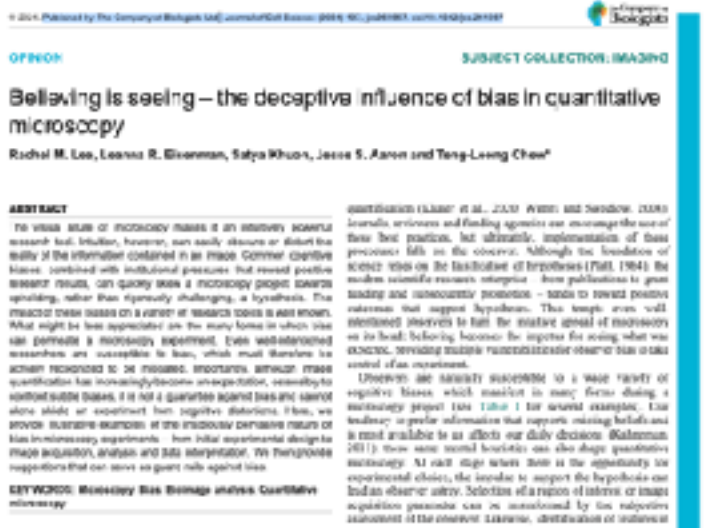


Which disk is larger?



Cognitive Bias

Clustering illusion	Seeing groups in time or space as significant when they are random (Gilovich et al., 1985).
Color perception	Illusions due to misleading perception of colors (Kovesi, 2015 preprint; Taylor et al., 2017; Wong, 2010)
Confirmation bias	Favoring information that supports existing beliefs (Mynatt et al., 1977; Nuzzo, 2015).
Congruence bias	Not testing alternative hypotheses for the observed data (Luckhoff, 2021; Wason, 1960).
Contrast effect	Over- or under-estimating a feature based on spatial-temporal surroundings (White, 1979).
Frequency illusion	Believing something just learned is more prevalent than it is (Bos et al., 2020)
Illusory correlation	Seeing a relationship where there is no underlying correlation (Chapman, 1967).
Pareidolia	Seeing patterns that do not exist (Foye et al., 2021; Voss et al., 2012).
Publication bias	Withholding negative results from publication (Joober et al., 2012).
Recency bias	Giving greater weight to more recent observations (Arnold et al., 2000).
Selection bias	Focusing on a sample that is not representative of the population (Jost and Waters, 2019).
Survivorship bias	Overlooking data that does not survive a selection process (B. Casselman 2016).
Weber–Fechner laws	Perception of a change is relative to the size of the changing feature (Dehaene, 2003).



**Believing is seeing –
the deceptive
influence of bias in
quantitative
microscopy**
R. M. Lee et al.
Journal of Cell
Science 2024

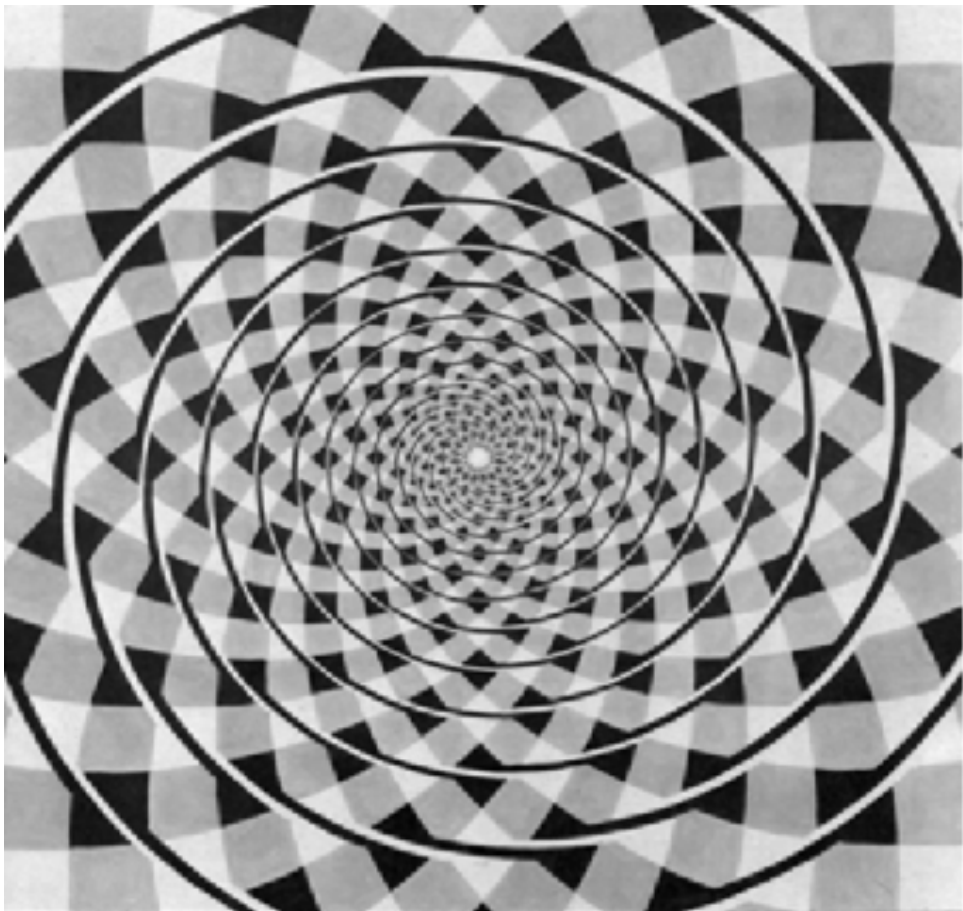
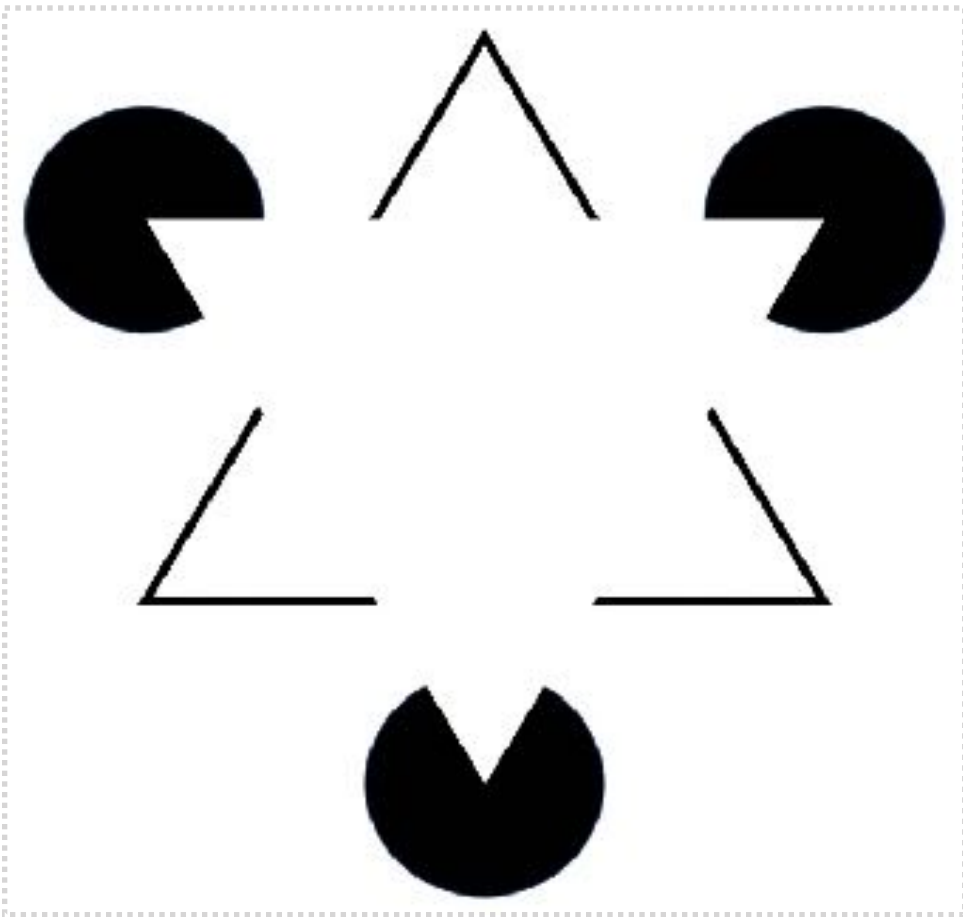
Share your experience



Seeing is Believing

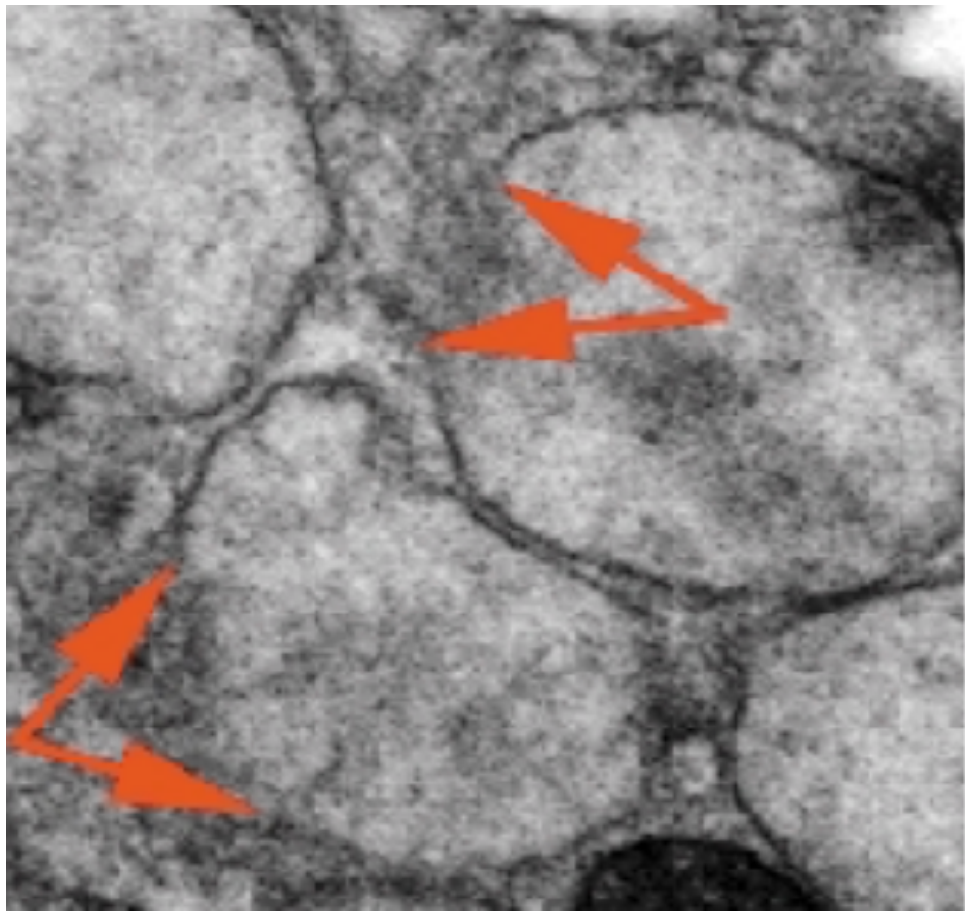
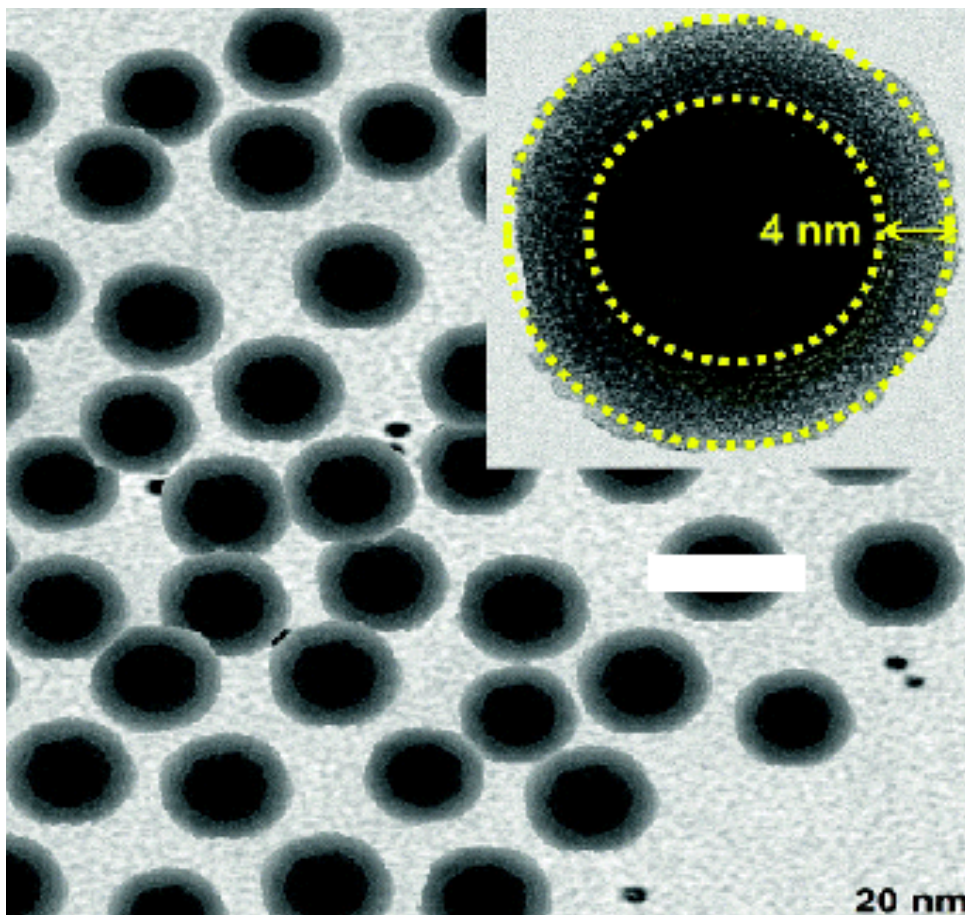
SEEING IS BELIEVING, QUANTIFYING IS SCIENCE

Hallucination



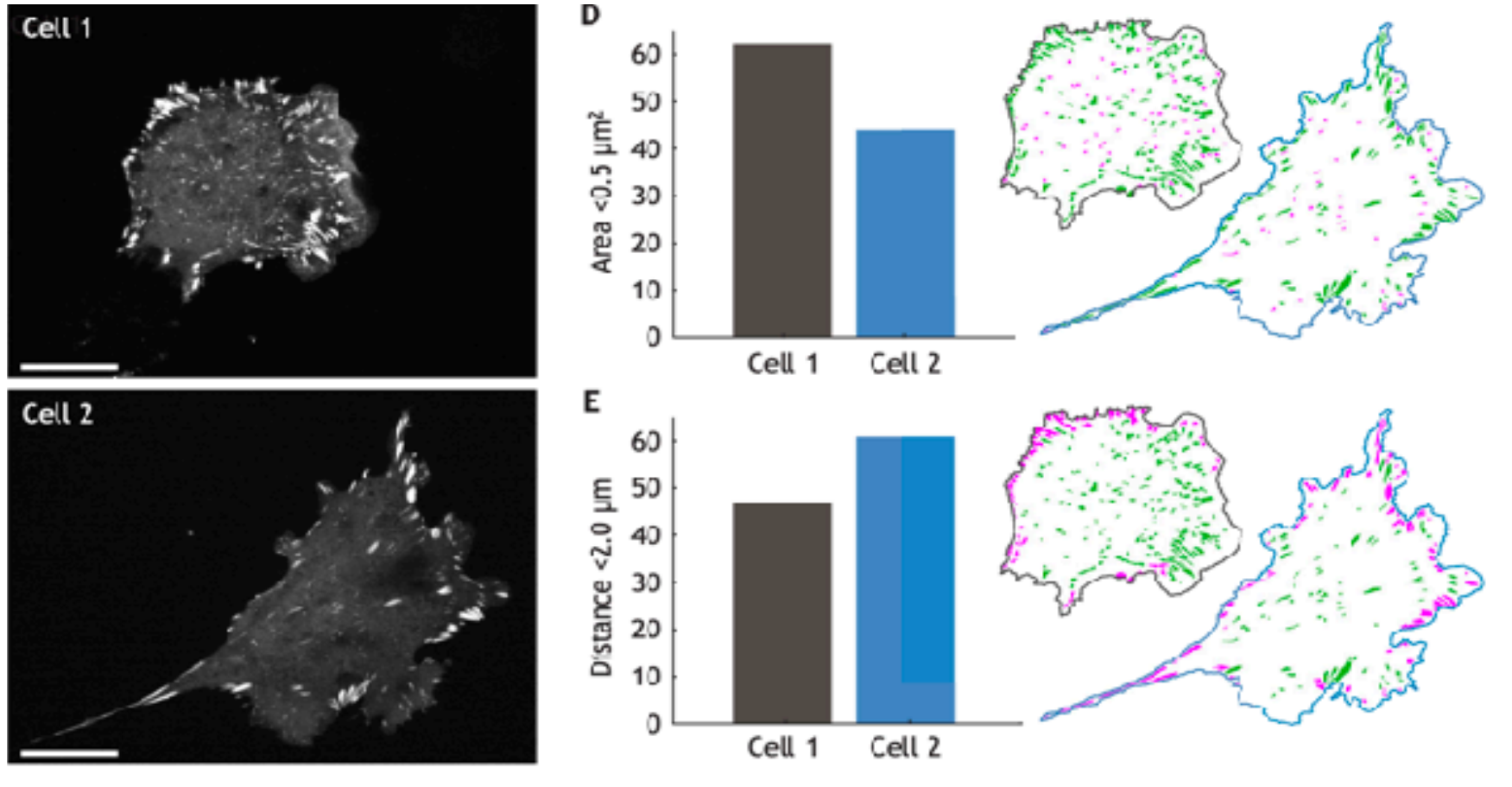
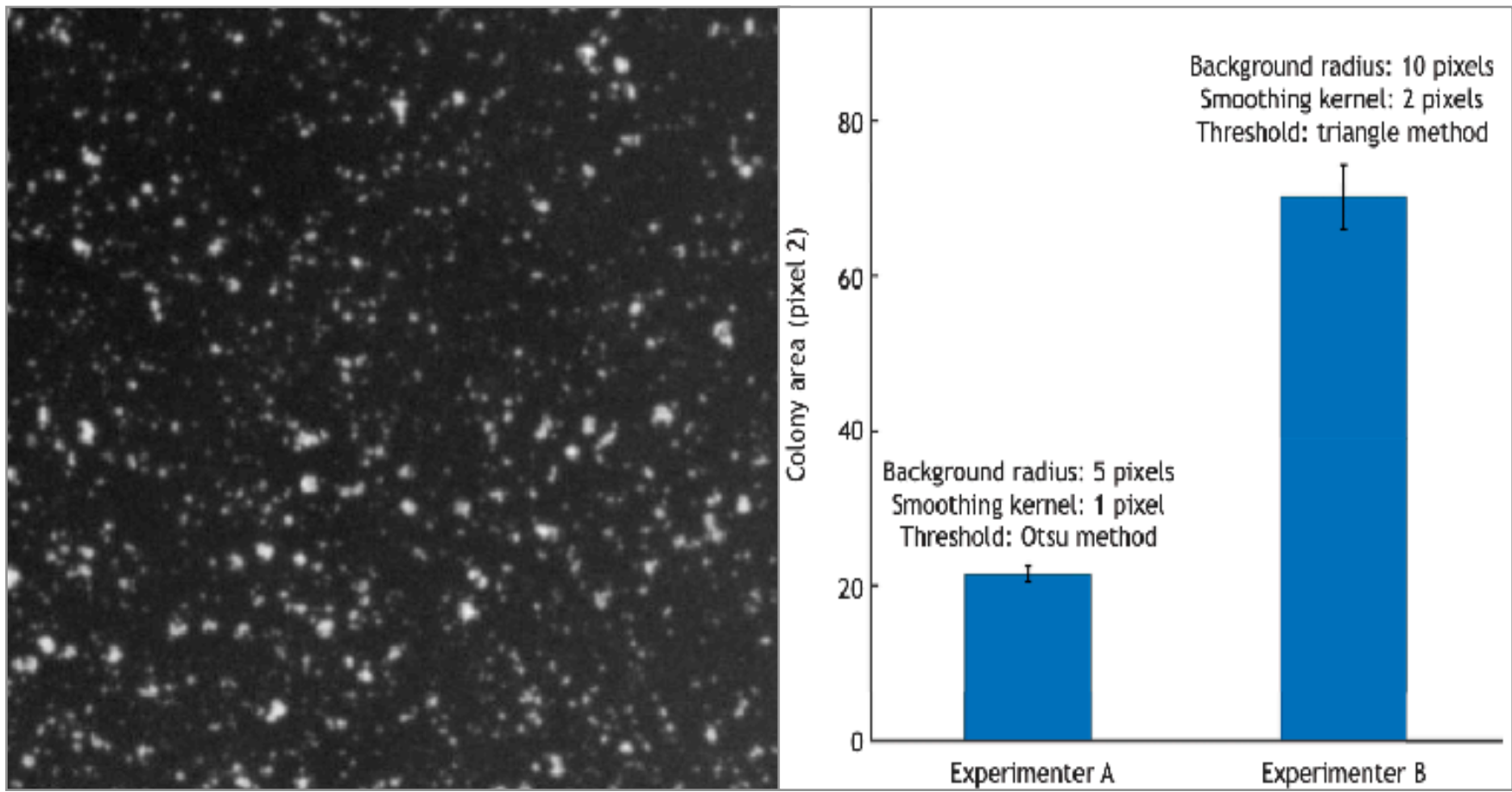
Seeing the scientific image, John C. Russ, www.drjohnruss.com

Interpretation



Seeing is believing? Alison J. North, J. of Cell Biology 2006

Quantitative Analysis



A guide to accurate reporting in digital image processing
J. Aaron and T.-L. Chew, J. of Cell Science 2024

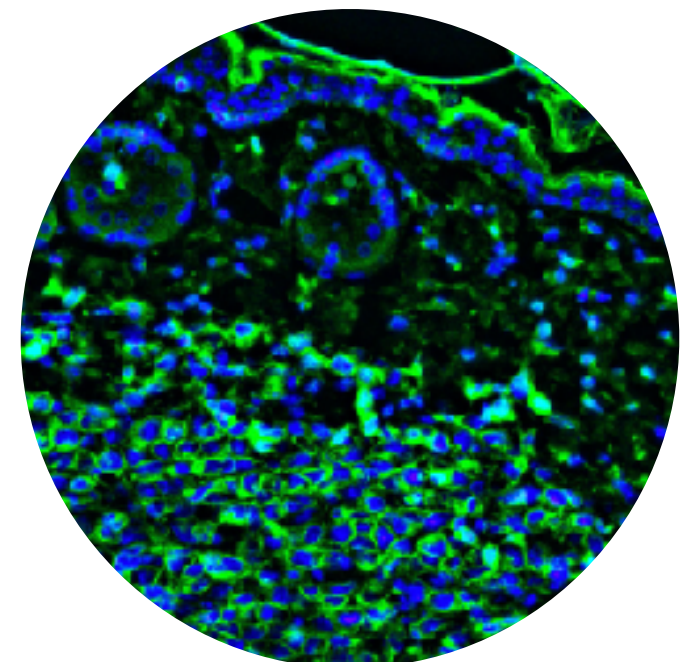


Ethics in Imaging

■ Data Integrity

Scientific Design of Training Dataset

- Proper curation annotation, validation
- Transparency in sourcing and usage
- Mitigation of bias perpetuation
- Promoting data integrity and openness



■ Ethics

Awareness of Impacts and Limitations

- Ethical compliance
- Clear code of conduct
- Intellectual property
- Prevention of AI manipulation



■ Education

Building Expertise in AI & Image Analysis

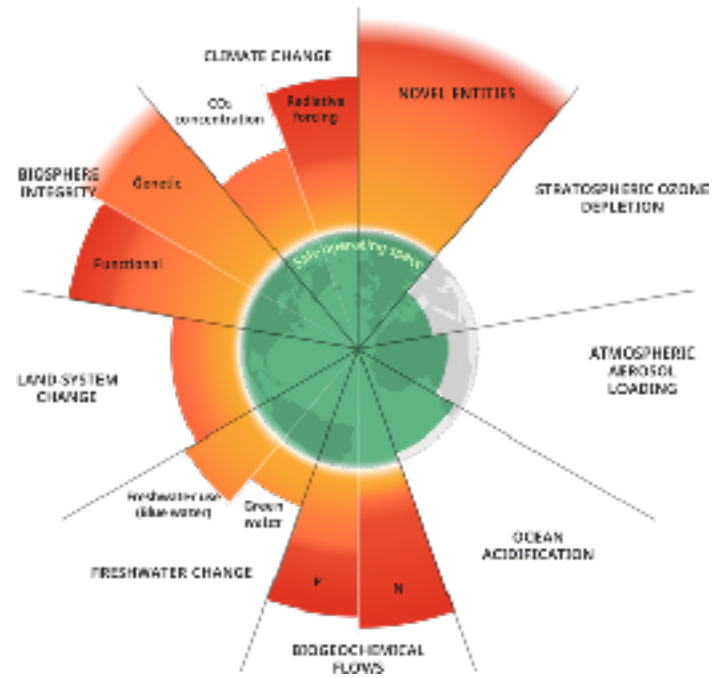
- Strong AI & imaging skills
- Engineering principles
- Share knowledge



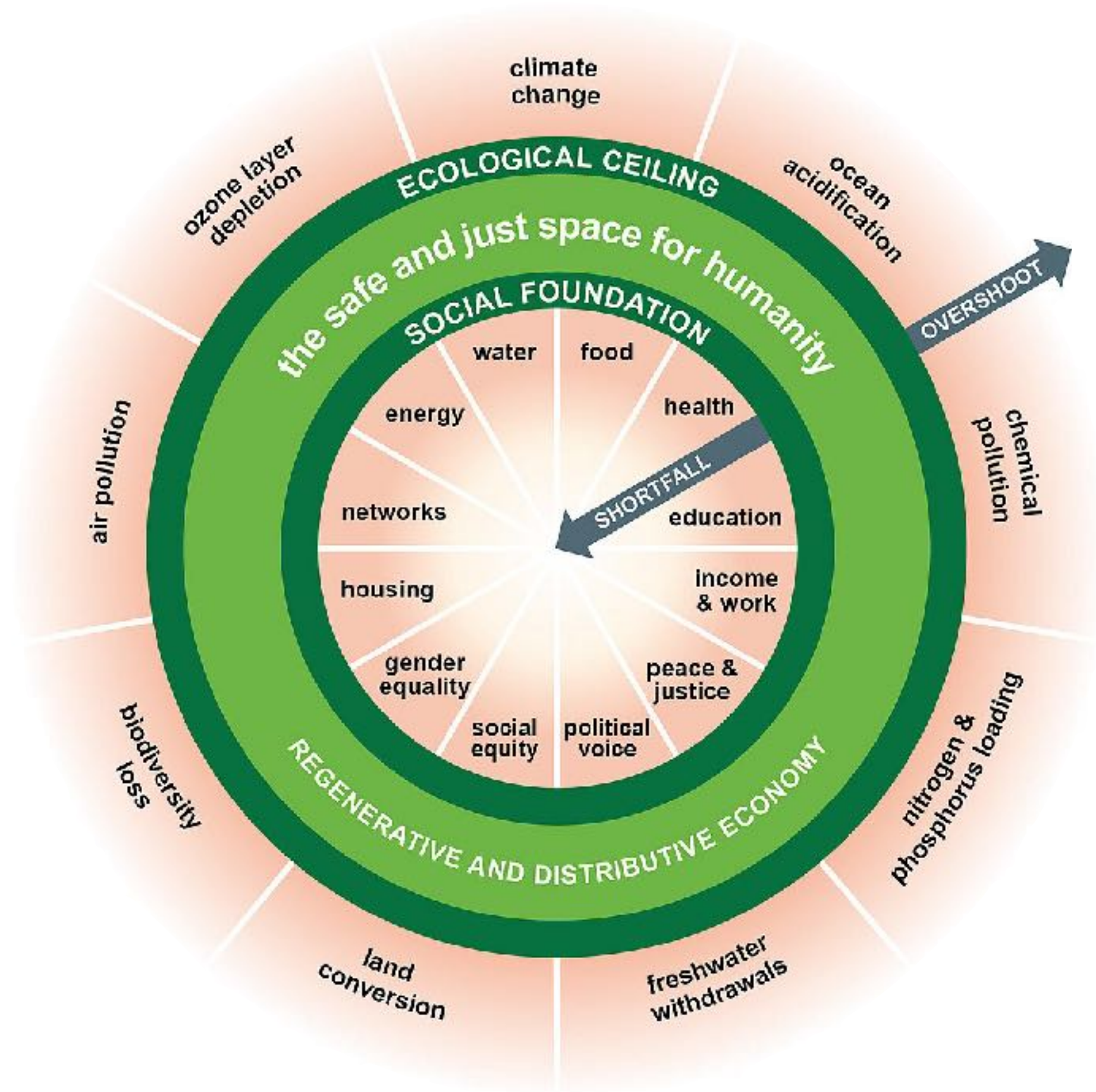
■ Responsibility

Societal and Environmental

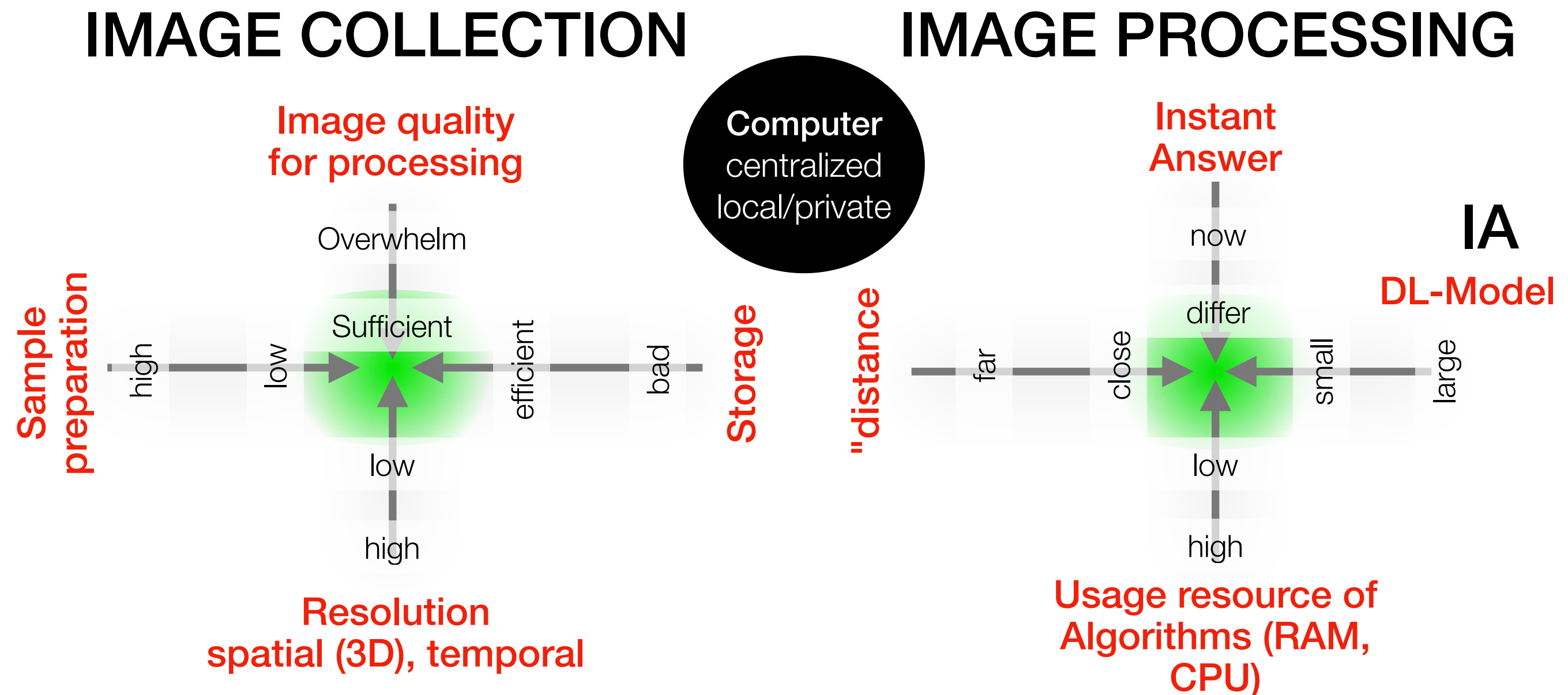
- Preventing of resource overshoot
- Reducing the ecological footprint
- Thinking as lifespan cycle
- Fair working conditions for annotators
- Sustainability & EGS (planetary boundaries)



👁 Sufficiency in Image Science



Create the doughnut of image science



The diagram was developed by [University of Oxford](#) economist [Kate Raworth](#) in her 2012 [Oxfam](#) paper *A Safe and Just Space for Humanity* and elaborated upon in her 2017 book *Doughnut Economics: Seven Ways to Think Like a 21st-Century Economist* and paper.^[3]



Digital Footprint

One iPhone requires 46 elements



iPhone 14

Footprint of 61Kg of CO₂e

Apple's Product Environmental Report

- 79% Sourcing & Manufacturing
- 2% Transportation
- 18% Usage
- 1% End of Life

Product's life cycle (3-4 years).

Greenly estimates that Apple is missing 80% of CO₂e

iPhone 14 should be 313 Kg CO₂e

- Use of carbon credits
- Lack of transparency
- Frequent updates
- Recent planned obsolescence
- Saving costs under the disguise of sustainability

More and more ...

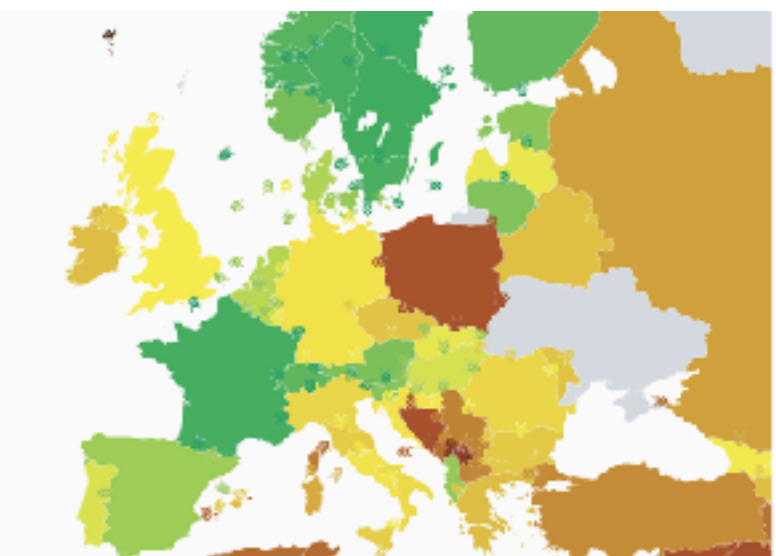
iPhone 6S Plus 63 Kg

iPhone 14 Pro Max 1TB 124 Kg

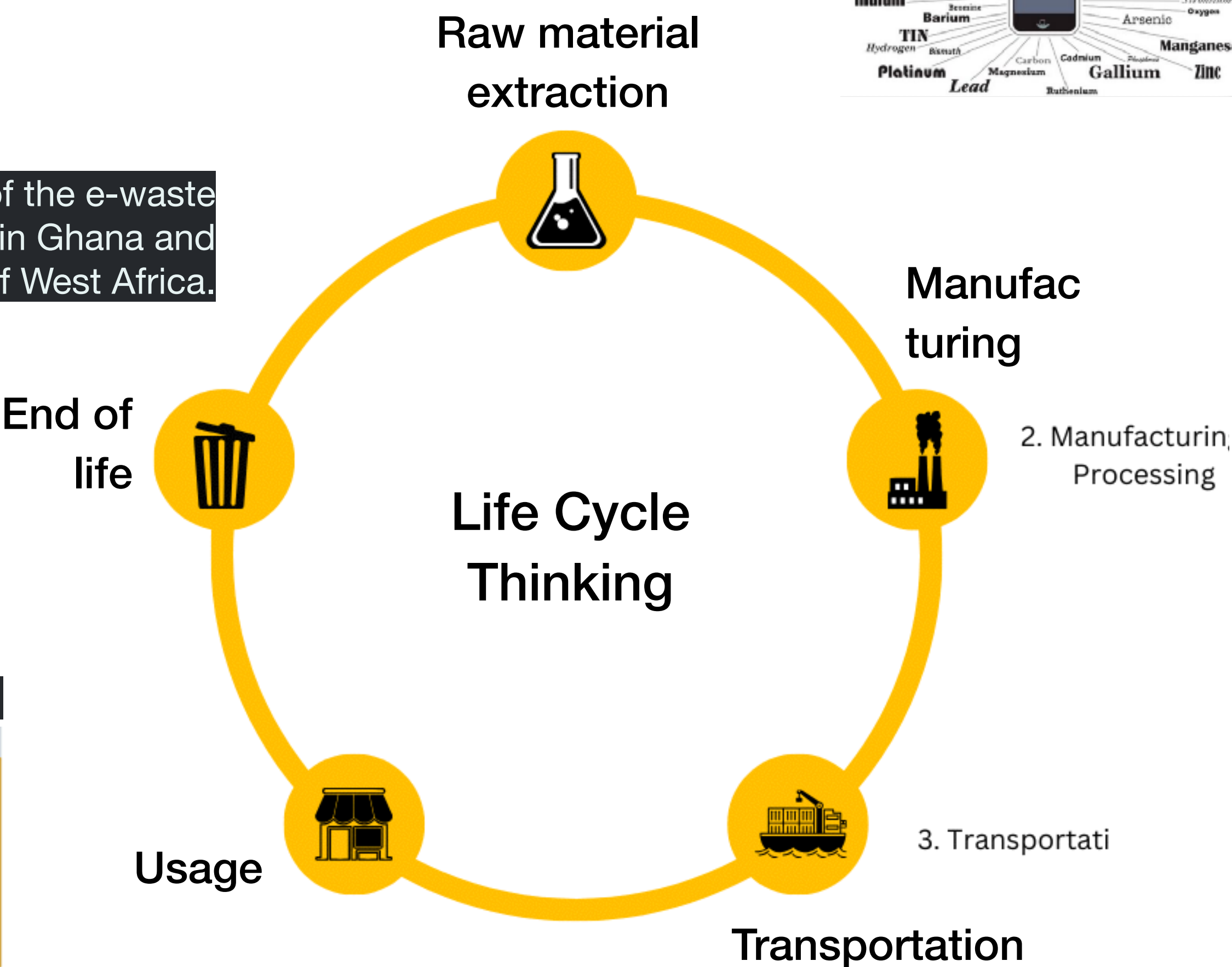
<https://www.shop-without-plastic.com/blogs/greenwashing/does-apple-do-greenwashing-2024>

About 85% of the e-waste
dumped in Ghana and
other parts of West Africa.

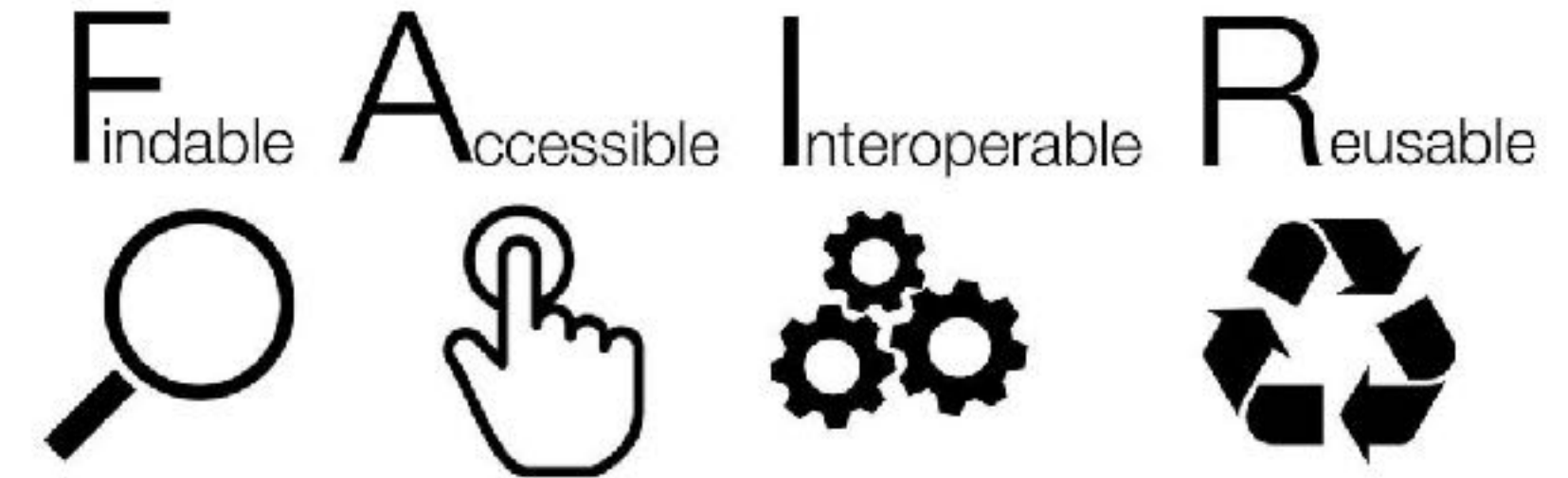
Where and when matters.



electricitymaps.com



How to Practice Open Science



Code

Open-source, versioning, test

→ Git

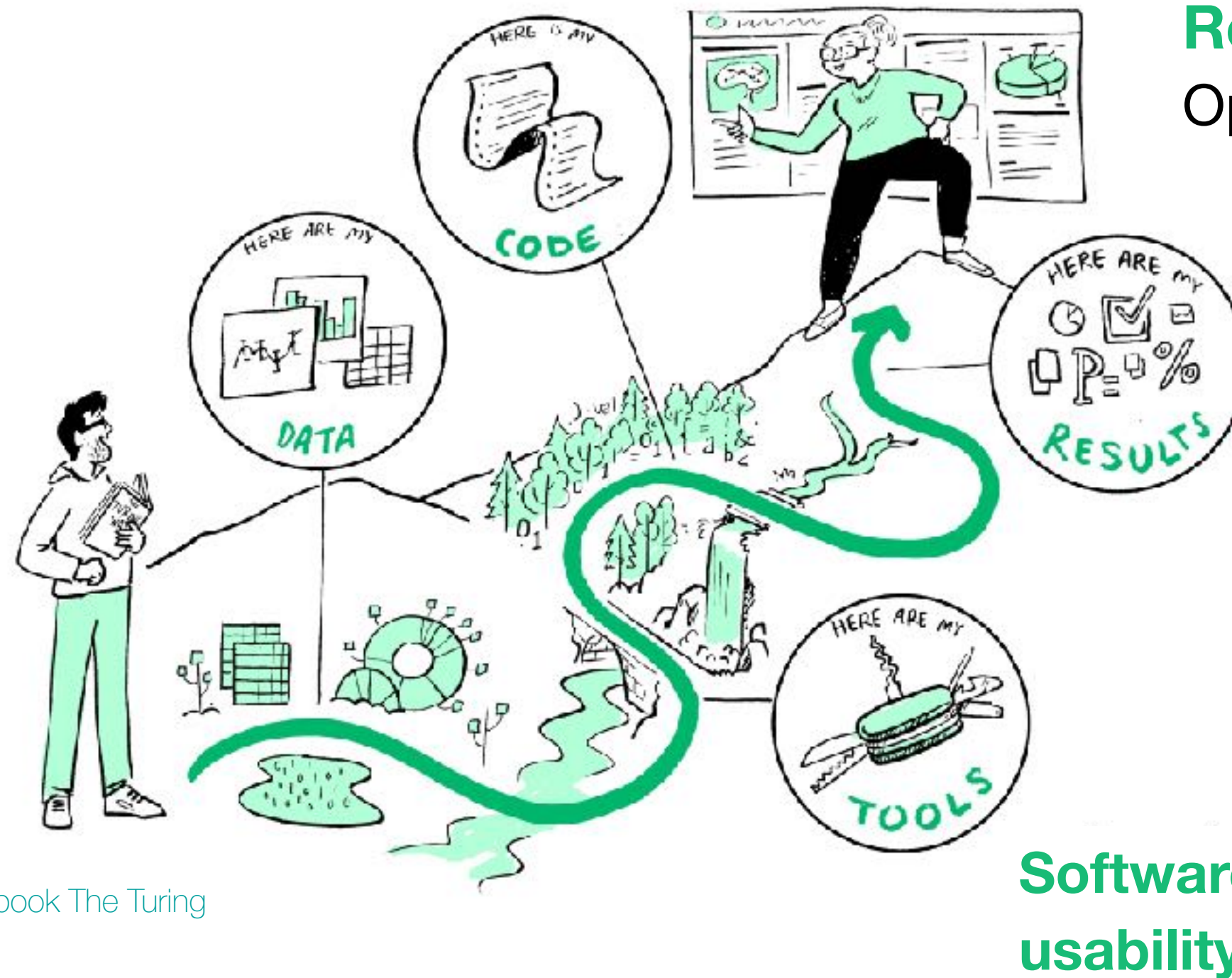
Data

Accessible
Meta-data

→ Zenodo

Results

Open Access
publication



Software
usability

Beatriz Serrano-Solano

Pproject can be found at [FAIR
Image Analysis Across
Sciences](#).

Article written by

Bioimage informatics:
usability is essential

*P. Paul-Gilloteaux, PLoS,
2023*

**A call for bioimaging
software usability**

Carpenter A, Nat. Meth. 2012

Why Procrastination?